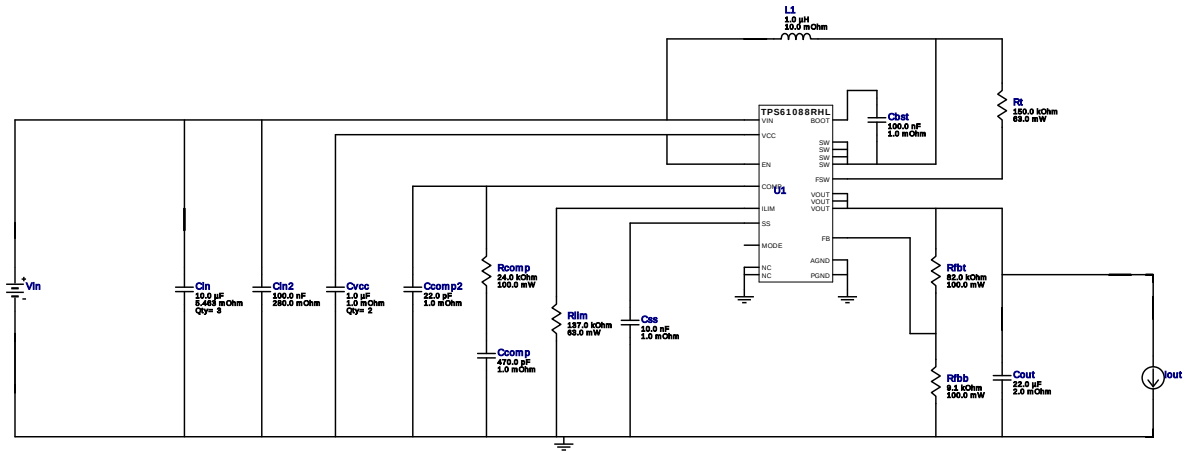
WEBENCH® Design Report

VinMin = 6.0V
VinMax = 9.0V
Vout = 12.0V
Iout = 2.0A

Device = TPS61088RHLR
Topology = Boost
Created = 2026-02-05 13:05:18.704
BOM Cost = $1.81
BOM Count = 18
Total Pd = 2.1W

Design : 15 TPS61088RHLR
TPS61088RHLR 6V-9V to 12.00V @ 2A

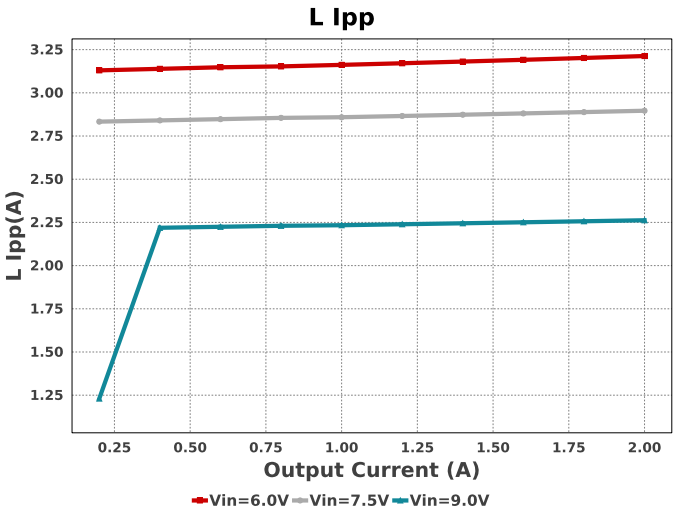


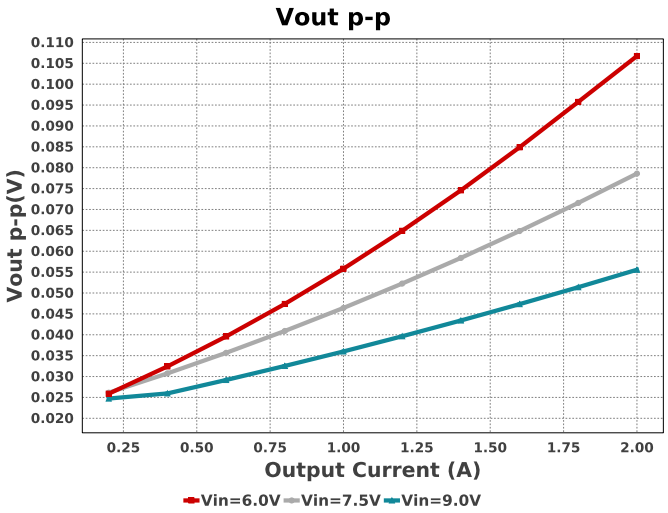
Electrical BOM

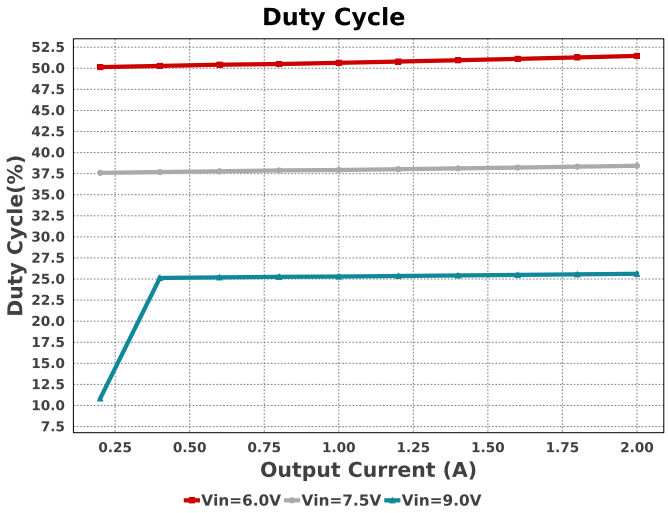
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cbst	MuRata	GRM155R71C104KA88D Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm²
Ccomp	MuRata	GRM033R71C471KA01D Series= X7R	Cap= 470.0 pF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0201 2 mm²
Ccomp2	AVX	02013A220GAT2A Series= C0G/NP0	Cap= 22.0 pF ESR= 1.0 mOhm VDC= 25.0 V IRMS= 0.0 A	1	$0.01	0201 2 mm²
Cin	Taiyo Yuden	MSAST21GBB5106MTNA01 Series= X5R	Cap= 10.0 uF ESR= 5.463 mOhm VDC= 25.0 V IRMS= 2.36768 A	3	$0.04	0805 7 mm²
Cin2	AVX	08053C104KAT2A Series= X7R	Cap= 100.0 nF ESR= 280.0 mOhm VDC= 25.0 V IRMS= 0.0 A	1	$0.01	0805 7 mm²
Cout	MuRata	GRM32ER61E226KE15L Series= X5R	Cap= 22.0 uF ESR= 2.0 mOhm VDC= 25.0 V IRMS= 3.67 A	1	$0.23	1210 15 mm²
Css	MuRata	GRM155R71C103KA01D Series= X7R	Cap= 10.0 nF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm²
Cvcc	Taiyo Yuden	EMK107B7105KA-T Series= X7R	Cap= 1.0 uF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	2	$0.01	0603 5 mm²
L1	Bourns	SRN8040-1R0Y	L= 1.0 µH 10.0 mOhm	1	$0.33	SRN8040 100 mm²

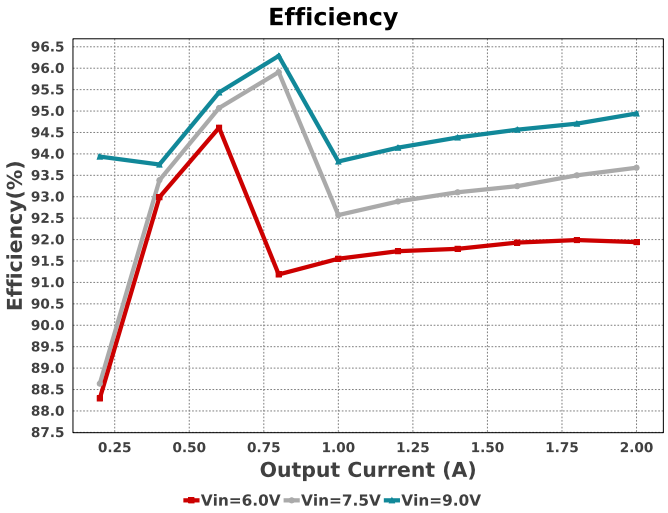
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Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Rcomp	Yageo	RC0603FR-0724KL Series= ?	Res= 24.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	$0.01	0603 5 mm²
Rfbb	Yageo	RC0603FR-079K1L Series= ?	Res= 9.1 kOhm Power= 100.0 mW Tolerance= 1.0%	1	$0.01	0603 5 mm²
Rfbt	Yageo	RC0603FR-0782KL Series= ?	Res= 82.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	$0.01	0603 5 mm²
Rlim	Vishay-Dale	CRCW0402137KFKED Series= CRCW..e3	Res= 137.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
Rt	Vishay-Dale	CRCW0402150KFKED Series= CRCW..e3	Res= 150.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
U1	Texas Instruments	TPS61088RHLR	Switcher	1	$1.01	RHL0020A 25 mm²

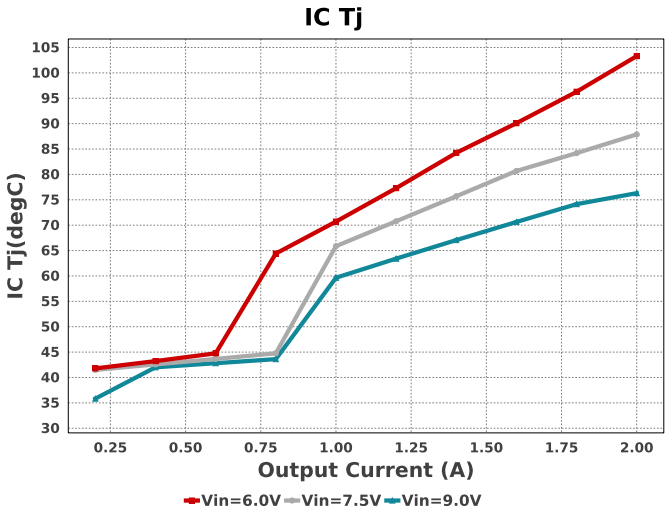


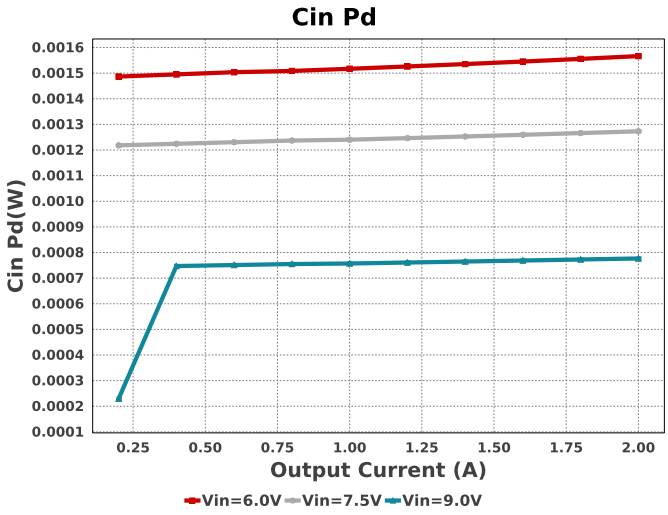


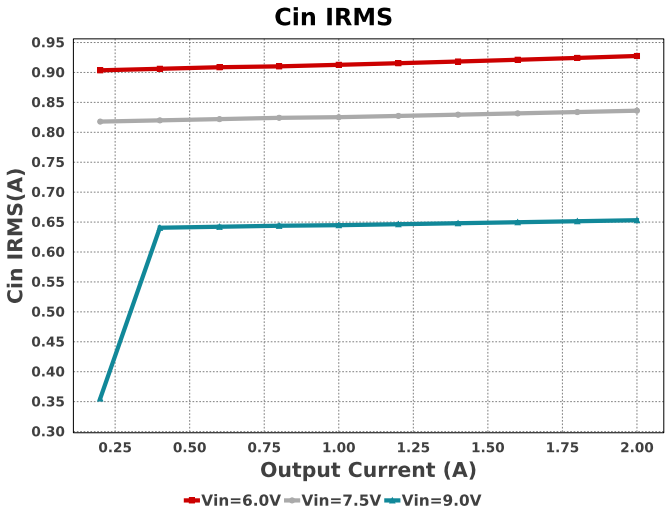


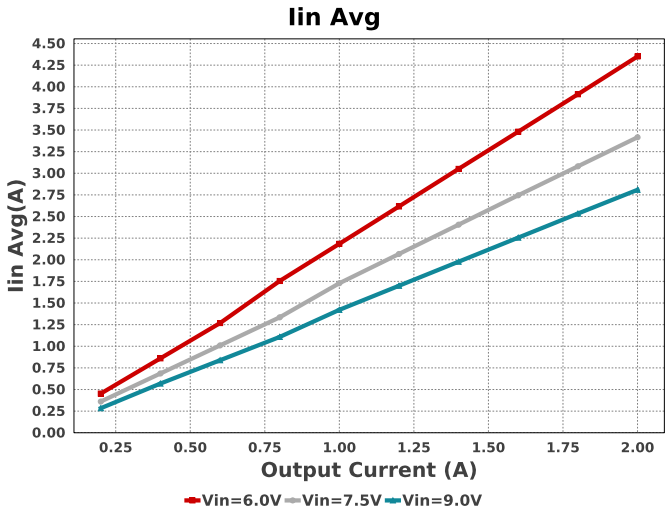


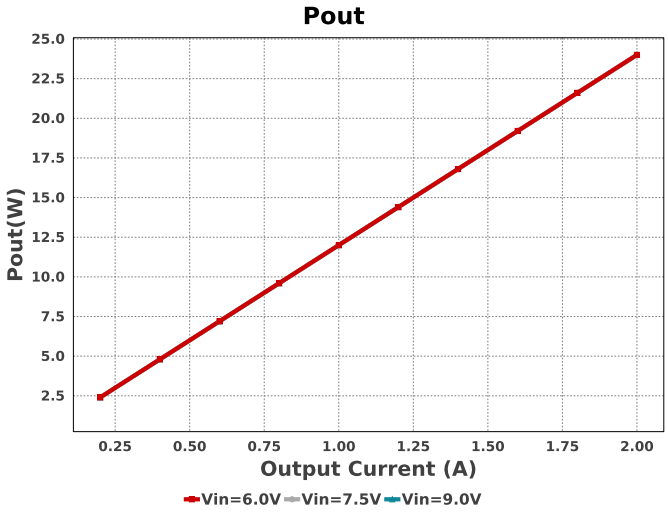
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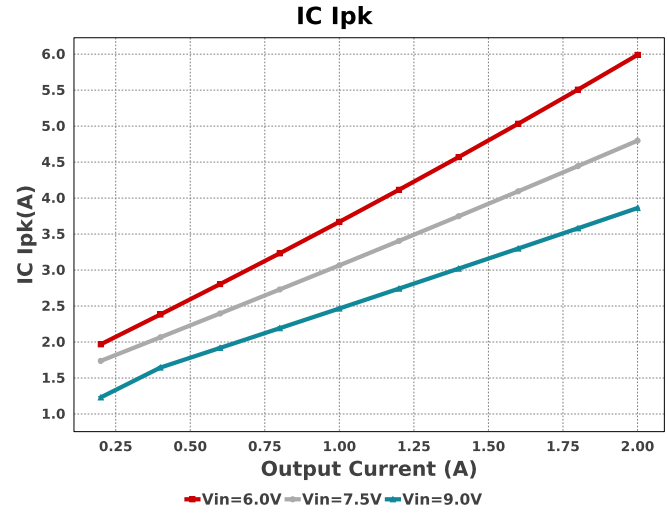




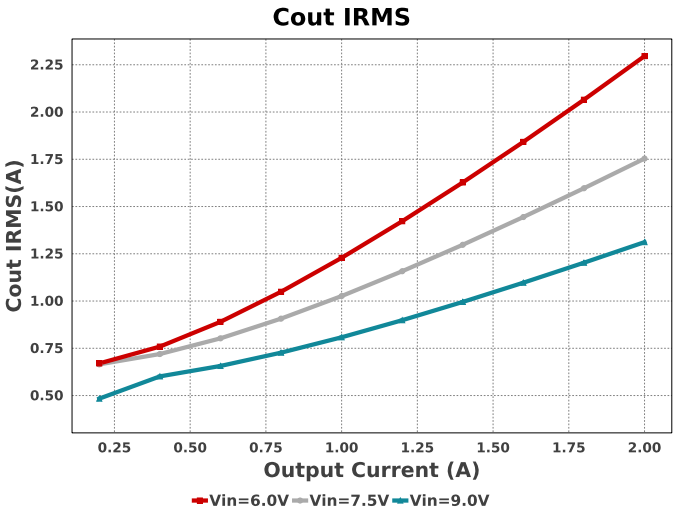


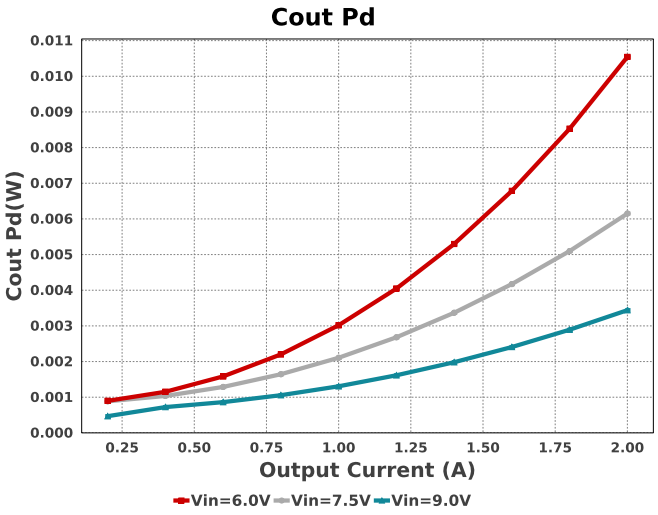


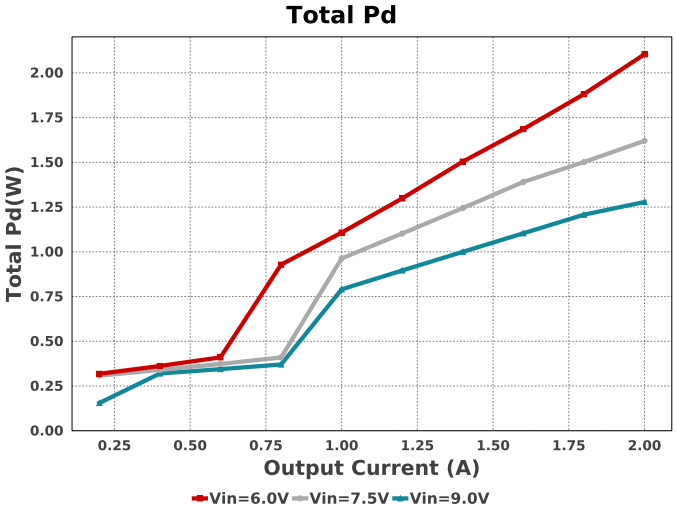


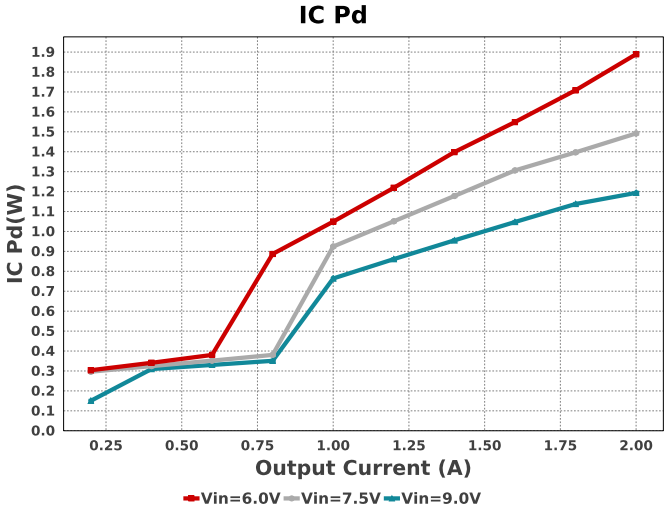


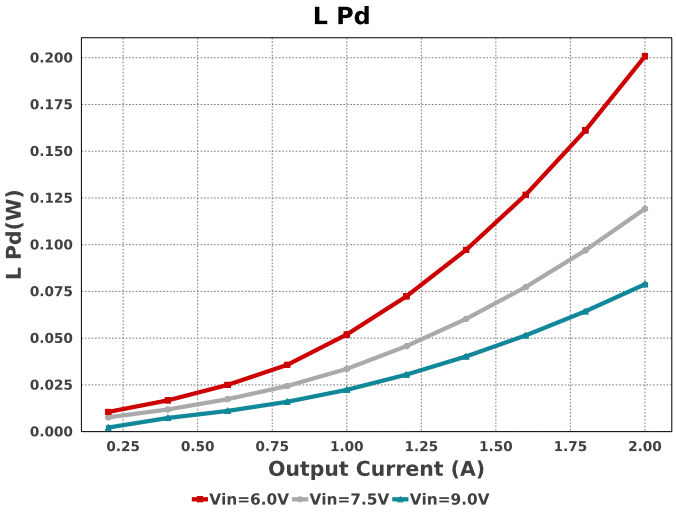
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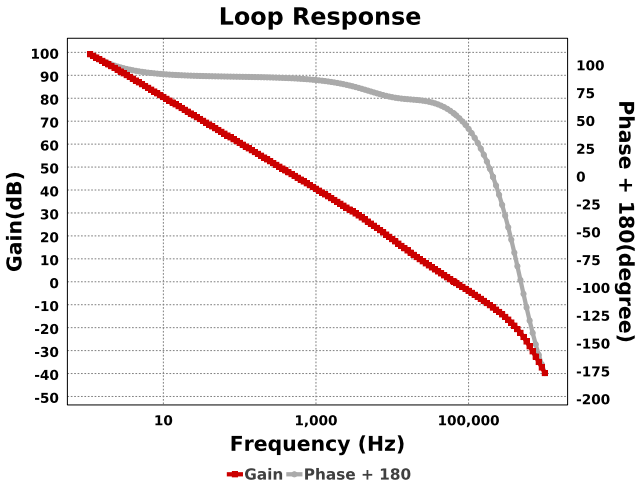




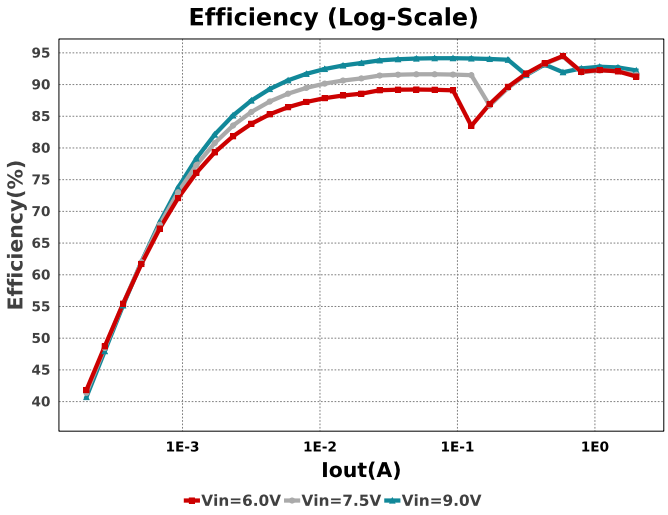








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Operating Values

#	Name	Value	Category	Description
1.	BOM Count	18		Total Design BOM count
2.	Total BOM	$1.812		Total BOM Cost
3.	Cin IRMS	927.499 mA	Capacitor	Input capacitor RMS ripple current
4.	Cin Pd	1.566 mW	Capacitor	Input capacitor power dissipation
5.	Cout IRMS	2.296 A	Capacitor	Output capacitor RMS ripple current
6.	Cout Pd	10.542 mW	Capacitor	Output capacitor power dissipation
7.	IC Ipk	5.99 A	IC	Peak switch current in IC
8.	IC Pd	1.889 W	IC	IC power dissipation
9.	IC Tj	103.307 degC	IC	IC junction temperature
10.	ICThetaJA	38.8 degC/W	IC	IC junction-to-ambient thermal resistance
11.	Iin Avg	4.351 A	IC	Average input current
12.	L Ipp	3.213 A	Inductor	Peak-to-peak inductor ripple current
13.	L Pd	200.77 mW	Inductor	Inductor power dissipation
14.	Cin Pd	1.566 mW	Power	Input capacitor power dissipation
15.	Cout Pd	10.542 mW	Power	Output capacitor power dissipation
16.	IC Pd	1.889 W	Power	IC power dissipation
17.	L Pd	200.77 mW	Power	Inductor power dissipation
18.	Total Pd	2.104 W	Power	Total Power Dissipation
19.	Cross Freq	64.623 kHz	System Information	Bode plot crossover frequency
20.	Duty Cycle	51.465 %	System Information	Duty cycle
21.	Efficiency	91.941 %	System Information	Steady state efficiency
22.	FootPrint	206.0 mm^2	System Information	Total Foot Print Area of BOM components
23.	Frequency	961.076 kHz	System Information	Switching frequency
24.	Gain Marg	-10.895 dB	System Information	Bode Plot Gain Margin
25.	Iout	2.0 A	System Information	Iout operating point
26.	Low Freq Gain	99.607 dB	System Information	Gain at 1Hz
27.	Mode	BOOST CCM	System Information	PWM/PFM Mode
28.	Phase Marg	55.738 deg	System Information	Bode Plot Phase Margin
29.	Pout	24.0 W	System Information	Total output power
30.	Vin	6.0 V	System Information	Vin operating point
31.	Vout Actual	12.053 V	System Information	Vout Actual calculated based on selected voltage divider resistors
32.	Vout Tolerance	4.482 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
33.	Vout p-p	106.738 mV	System Information	Peak-to-peak output ripple voltage

Design Inputs

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Name	Value	Description
Iout	2.0	Maximum Output Current
SoftStart	2.41 ms	Soft Start Time (ms)
VinMax	9.0	Maximum input voltage
VinMin	6.0	Minimum input voltage
Vout	12.0	Output Voltage
base_pn	TPS61088	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature
UserFsw	961.0 k	Customer Selected Frequency

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WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of Cin and Cout, and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

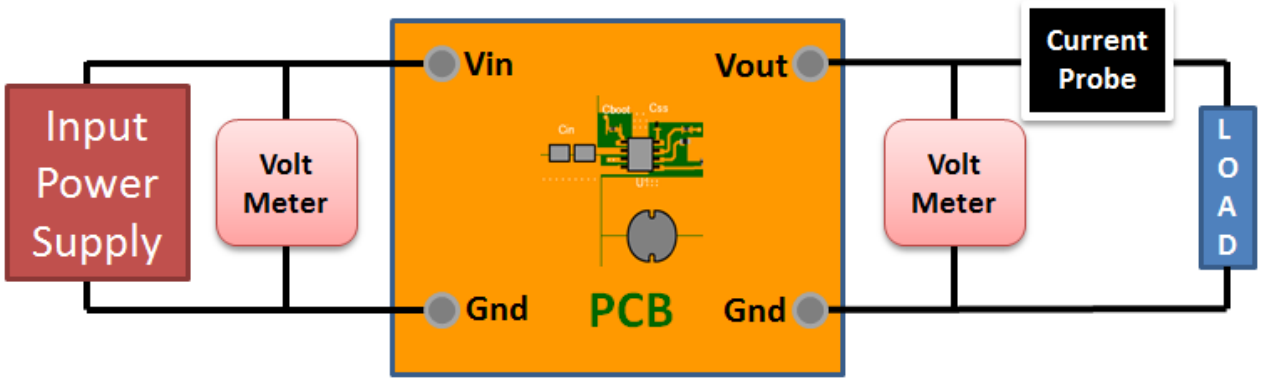
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab town to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 6.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to Vin and GND. Connect a digital volt meter and a load if needed to set the minimum Iout of the design from Vout and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between Vin and GND, a load is connected between Vout and GND and a current meter is connected in series between Vout and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : A334E1D4DAA32D88[v1]
2. **TPS61088** Product Folder : http://www.ti.com/product/TPS61088 : contains the data sheet and other resources.

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